

Low Vision Assessment in Virtual Reality (LVAVR): Spatial Computing in Virtual Reality and Real-World

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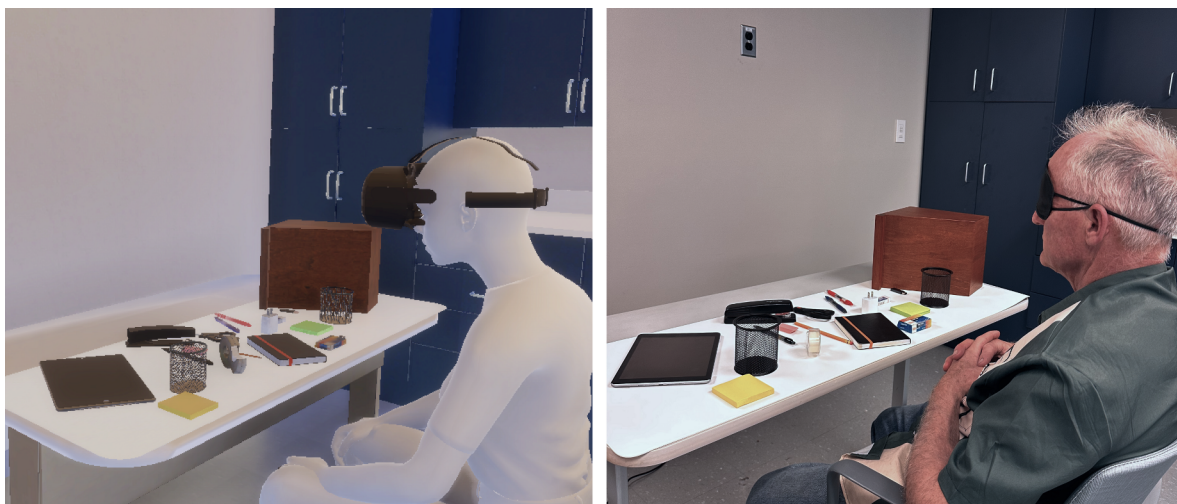


Figure 1: LVAVR VR and RW activity example (cluttered scene).

ABSTRACT

The Low Vision Assessment in Virtual Reality project aims to create performance assessments of activities of daily living for a broad low vision population in virtual reality. In this study, our goal is to validate representative virtual reality activities against matching real-world activities to create standardized activities of daily living in virtual reality. Experiments were conducted in a rehabilitation space with a kitchen and living room, which was recreated in virtual reality with photorealistic rendering. We created 25 activities in Unity based on our previously developed visual functioning questionnaire and performed a validation study.

Index Terms: Virtual Reality, Real World, Low Vision.

1 INTRODUCTION

The Low Vision Assessment in Virtual Reality project aims to create and validate performance measures for functional assessment of a broad low vision (LV) population (visual acuity between 20/50 and 20/800) in virtual reality (VR). Functional vision can be assessed using self-report (Visual Functioning Questionnaires (VFQ)) and performance measures (Standardized Activity outcomes). Performance measures take the form of accuracy and completion times

in activities of daily living (ADL). Our goal is to validate representative VR activities against matching real-world (RW) activities to create standardized ADLs. Our proposed visual performance assessment is representative of LV patients' vision use in daily life and covers five visual settings including self-care, technology, work, housekeeping, and community.

Due to limitations of performance measures and standardized ADLs for clinical application in the real world (e.g., test environmental conditions), we developed the same activities in RW and VR, to assess the feasibility of continuing the next phase of this research solely in VR. The activities in this study are selected from a previously developed Low Vision-Visual Function Questionnaire.

2 BACKGROUND

Previous research demonstrated the feasibility of functional vision assessment to an ultra-low vision and prosthetic vision population. Functional vision assessments for individuals with ultra-low vision (Visual Functioning Questionnaires and performance measures using activities in VR and RW) have been previously reported [2, 7, 9, 11, 12].

In order to remove limitations and biases of self-reported assessments, objective measures such as the user's physiological responses, task completion time and accuracy can be used to "corroborate" these subjective measures [8, 13, 17, 18].

In VR research, immersion refers to spatial immersion where the virtual environment and the stimuli are comparable to the real world [5, 15]. Comparison of RW and VR has been extensively studied, and some results indicate no significant difference between the RW and VR in perceptual accuracy [6, 10, 19]. However, color and lightness constancy have shown a significant difference in RW compared to VR [14, 16].

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3 METHODS

We recruited LV patients based on their visual acuity (visual acuity between 20/50 and 20/800). In the pilot phase, we conducted experiments with 11 activities in RW and their comparable activities in VR using a Varjo XR-4 head-mounted display among 7 LV patients and 5 control subjects (normally sighted). The second phase includes 25 activities in VR and RW with the addition of assistive technologies. All experiments were conducted in a rehabilitation space with a kitchen and living room. The VR experience was created in Unity using High-Definition Render Pipeline to maximize user immersion. We explored different methods to create our 3D space, including 3D Gaussian Splatting and 3D modeling, and chose to use 3D modeling due to graphics integration limitations while using the Varjo XR-4 in Unity. Lighting properties in RW were measured using an LS-150 Luminance Meter and an LM-81LX light meter and were used to assign lighting properties in Unity. The materials were photographed and recreated using Adobe 3D sampler [1]. We used the method of successive dichotomizations [3, 4] to select activities and estimate the item measures (IM). Standing activities include identifying kitchen ingredients (flour vs. sugar and flour vs. rice) and performing hand-eye coordination tasks (rotating stove knob). Seated activities include identifying familiar faces, lengthy and short reading tasks, and identifying objects in a cluttered scene. We analyzed the results using the Shapiro-Wilk test to verify normality, followed by a t-test or Mann-Whitney test depending on the outcome.

4 RESULTS

Patients' performances in VR and RW were strongly correlated across all tasks (Pearson correlation, $r = 0.93$, $p < 0.001$). Our statistical tests showed no significant difference in completion time among LV patients in most activities between RW and VR (t-test, $p > 0.05$). Only two activities indicated a significant difference between RW and VR: hand-eye coordination activity (rotating stove knob) and identifying kitchen ingredients (flour vs. rice) (t-test, $p < 0.05$). The most successful activities were short and lengthy readings, identifying familiar faces, identifying objects in a cluttered scene, and identifying spices. No significant differences were found between VFQ-derived IMs and task performances in VR (t-test, $p > 0.05$) and RW (t-test, $p > 0.05$). Control subjects showed no significant difference between VR and RW for reading tasks.

5 CONCLUSIONS

In this study, we introduced comparable activities in VR and RW for LV patients. Our results show that low-vision patient-reported outcome measures and objective visual function may not consistently align due to patient reliance on assistive technologies and subjective adaptation to visual impairment. These findings align with our current VR activities, which did not provide magnification and contrast enhancement/inversion. We will introduce the equivalent of patients' assistive technologies in VR in the next phase of this research. More hand tracking training should be done before the experiments. Activities requiring realistic material representation will be modified with improved rendering (identifying ingredients). We are currently recruiting more patients from all groups to improve the sensitivity comparing VR and RW activities.

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